

Newman-Penrose Formalism Calculations

Stephani et al., Equation (15.17_6)

Metric

$$ds^2 = \frac{2m}{r} du \otimes du - du \otimes dr - dr \otimes du + r^2 Y(u, r)^2 dx_1 \otimes dx_1 + r^2 x_1^2 Y(u, r)^2 dx_2 \otimes dx_2$$

Coordinates

$$(x^\mu) = (u, r, x_1, x_2)$$

Metric Functions

$$\Sigma = x_1$$

$$H = -\frac{2m}{r}$$

Null Tetrad

Covariant components

$$l = (1) du$$

$$n = \left(-\frac{m}{r}\right) du + (1) dr$$

$$m = \left(\frac{1}{2} \sqrt{2} r Y(u, r)\right) dx_1 + \left(\frac{1}{2} i \sqrt{2} r x_1 Y(u, r)\right) dx_2$$

$$\bar{m} = \left(\frac{1}{2} \sqrt{2} r Y(u, r)\right) dx_1 + \left(-\frac{1}{2} i \sqrt{2} r x_1 Y(u, r)\right) dx_2$$

Contravariant components

$$l = (-1) \, dr$$

$$n = (-1) \, du + \left(-\frac{m}{r}\right) \, dr$$

$$m = \left(\frac{\sqrt{2}}{2rY(u,r)}\right) \, dx_1 + \left(\frac{i\sqrt{2}}{2rx_1Y(u,r)}\right) \, dx_2$$

$$\overline{m} = \left(\frac{\sqrt{2}}{2rY(u,r)}\right) \, dx_1 + \left(-\frac{i\sqrt{2}}{2rx_1Y(u,r)}\right) \, dx_2$$

Directional Derivatives

$$D = -\partial_r$$

$$\Delta = -\partial_u - \frac{m}{r} \, \partial_r$$

$$\delta = \frac{\sqrt{2}}{2rY(u,r)} \, \partial_{x_1} + \frac{i\sqrt{2}}{2rx_1Y(u,r)} \, \partial_{x_2}$$

$$\bar{\delta} = \frac{\sqrt{2}}{2rY(u,r)} \, \partial_{x_1} - \frac{i\sqrt{2}}{2rx_1Y(u,r)} \, \partial_{x_2}$$

Spin Coefficients

$$\rho = \frac{\frac{\partial}{\partial r}Y(u,r)}{Y(u,r)} + \frac{1}{r}$$

$$\sigma = 0$$

$$\kappa = 0$$

$$\mu = -\frac{\frac{\partial}{\partial u}Y(u,r)}{Y(u,r)} - \frac{m\frac{\partial}{\partial r}Y(u,r)}{rY(u,r)} - \frac{m}{r^2}$$

$$\lambda = 0$$

$$\nu = 0$$

$$\alpha = -\frac{\sqrt{2}}{4rx_1Y(u,r)}$$

$$\beta = \frac{\sqrt{2}}{4rx_1Y(u,r)}$$

$$\tau = 0$$

$$\pi = 0$$

$$\epsilon = 0$$

$$\gamma = -\frac{m}{2r^2}$$

Ricci Tensor Components

$$\Phi_{00} = -\frac{\frac{\partial^2}{(\partial r)^2}Y(u,r)}{Y(u,r)} - \frac{2\frac{\partial}{\partial r}Y(u,r)}{rY(u,r)}$$

$$\Phi_{01} = 0$$

$$\Phi_{02} = 0$$

$$\Phi_{10} = 0$$

$$\Phi_{11} = \frac{\frac{\partial}{\partial u} Y(u, r) \frac{\partial}{\partial r} Y(u, r)}{2 Y(u, r)^2} + \frac{m \frac{\partial}{\partial r} Y(u, r)^2}{2 r Y(u, r)^2} + \frac{\frac{\partial}{\partial u} Y(u, r)}{2 r Y(u, r)} + \frac{m \frac{\partial}{\partial r} Y(u, r)}{r^2 Y(u, r)}$$

$$\Phi_{12} = 0$$

$$\Phi_{20} = 0$$

$$\Phi_{21} = 0$$

$$\Phi_{22} = -\frac{\frac{\partial^2}{(\partial u)^2} Y(u, r)}{Y(u, r)} - \frac{2 m \frac{\partial^2}{\partial u \partial r} Y(u, r)}{r Y(u, r)} - \frac{m^2 \frac{\partial^2}{(\partial r)^2} Y(u, r)}{r^2 Y(u, r)} - \frac{3 m \frac{\partial}{\partial u} Y(u, r)}{r^2 Y(u, r)} - \frac{2 m^2 \frac{\partial}{\partial r} Y(u, r)}{r^3 Y(u, r)}$$

$$\Lambda = \frac{\frac{\partial^2}{\partial u \partial r} Y(u, r)}{3 Y(u, r)} + \frac{\frac{\partial}{\partial u} Y(u, r) \frac{\partial}{\partial r} Y(u, r)}{6 Y(u, r)^2} + \frac{m \frac{\partial}{\partial r} Y(u, r)^2}{6 r Y(u, r)^2} + \frac{m \frac{\partial^2}{(\partial r)^2} Y(u, r)}{3 r Y(u, r)} + \frac{\frac{\partial}{\partial u} Y(u, r)}{2 r Y(u, r)} + \frac{2 m \frac{\partial}{\partial r} Y(u, r)}{3 r^2 Y(u, r)}$$

Weyl Tensor Components

$$\Psi_0 = 0$$

$$\Psi_1 = 0$$

$$\Psi_2 = \frac{\frac{\partial^2}{\partial u \partial r} Y(u, r)}{3 Y(u, r)} - \frac{\frac{\partial}{\partial u} Y(u, r) \frac{\partial}{\partial r} Y(u, r)}{3 Y(u, r)^2} - \frac{m \frac{\partial}{\partial r} Y(u, r)^2}{3 r Y(u, r)^2} + \frac{m \frac{\partial^2}{(\partial r)^2} Y(u, r)}{3 r Y(u, r)} - \frac{m \frac{\partial}{\partial r} Y(u, r)}{3 r^2 Y(u, r)} - \frac{m}{r^3}$$

$$\Psi_3 = 0$$

$$\Psi_4 = 0$$

Newman-Penrose Equations

Equations are vanished.

Bianchi Identities

Equations are vanished.

Petrov Type

Type "D"